

BIG DATA ANALYTICS – CASE STUDY

MEDTRACK – HADOOP MAPREDUCE ANALYTICS

TEAM - 3

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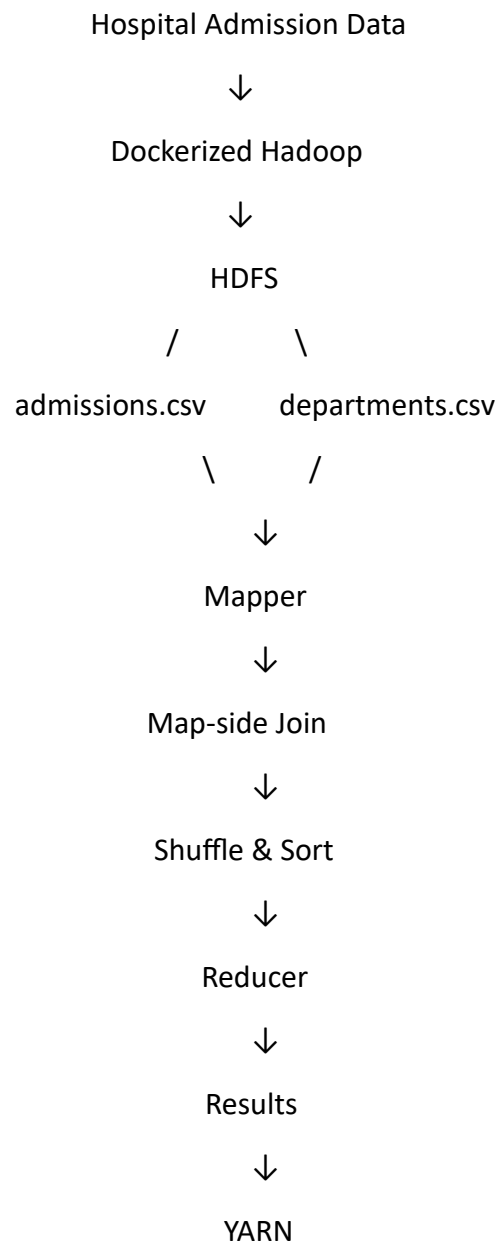
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Introduction

MedTrack is a healthcare data analytics project implemented using Hadoop MapReduce. The project processes hospital admission data stored in HDFS and performs five analytical tasks related to department admissions, patient stay, monthly turnover, department heads, and monthly admission trends. The MapReduce jobs are executed in a Dockerized Hadoop environment using HDFS and YARN.

System Architecture



The admission and department datasets are stored in HDFS. The Mapper processes admission records and, where required, uses the department dataset through a Map-side Join. Hadoop performs Shuffle and Sort to group intermediate results. The Reducer performs

the required aggregation and generates the final output. YARN manages the execution of the MapReduce jobs.

Hadoop Environment

Component	Technology
Operating System	Ubuntu/WSL
Container Platform	Docker
Processing	Hadoop MapReduce
Storage	HDFS
Resource Management	YARN
Hadoop Version	Hadoop 3.2.1

HDFS Data Ingestion

```
hdfs dfs -ls /medtrack/input
```

OUTPUT:

```
root@685e756c510d:/tmp# hdfs dfs -ls /medtrack/input
Found 2 items
-rw-r--r--  3 root supergroup      10976 2026-09-21 16:16 /medtrack/input/admissions.csv
-rw-r--r--  3 root supergroup       359 2026-09-21 16:21 /medtrack/input/departments.csv
```

Problem 1 – Total Admissions per Department

Objective

Calculate the total number of admissions/patients for each department.

Approach

The Mapper uses a Map-side Join to obtain the department name from the department dataset and emits the department name with a count of 1. The Reducer sums the counts for each department.

OUTPUT:

```
PS C:\Users\B Lalith kumar\Desktop\MedTrack-Problem1> docker exec namenode hdfs dfs -cat /output/medtrack-problem1/part-r-00000
WARNING: log4j.properties is not found. HADOOP_CONF_DIR may be incomplete.
Cardiology      4
Neurology       3
Orthopedics     3
PS C:\Users\B Lalith kumar\Desktop\MedTrack-Problem1>
```

Problem 2 – Average Length of Stay

Objective

Calculate the average length of stay for each department.

Approach

The Mapper calculates the length of stay using the admission and discharge dates and emits the department with the calculated stay. The Reducer aggregates the stay values and calculates the average.

OUTPUT:

```
root@165f0de63dbe:~/medtrack# hdfs dfs -cat /medtrack/output_p2/part-r-00000
2026-09-27 17:10:13,409 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteHostTrusted = false
Cardiology      : 4.66 days
Dermatology     : 5.00 days
ENT             : 4.05 days
General Medicine : 4.60 days
Gynecology      : 4.05 days
Neurology       : 4.43 days
Oncology        : 4.68 days
Orthopedics     : 4.29 days
Pediatrics      : 4.60 days
Pulmonology     : 4.55 days
```

Problem 3 – Highest Patient Turnover in a Month

Objective

Find the department with the highest patient turnover for each month.

Approach

The Mapper extracts the admission month and department name. Hadoop groups records by month. The Reducer counts admissions for each department within each month and selects the department with the highest count.

OUTPUT:

```
root@685e756c510d:/tmp# hdfs dfs -cat /med
2026-09-21 16:49:20,076 INFO sasl.SaslData
2026-01 : General Medicine (8 patients)
2026-02 : Neurology (7 patients)
2026-03 : Cardiology (7 patients)
2026-04 : General Medicine (8 patients)
2026-05 : Neurology (7 patients)
2026-06 : General Medicine (6 patients)
root@685e756c510d:/tmp#
```

Problem 4 – Department Head with Most Admissions

Objective

Identify the department head overseeing the highest number of admissions.

Approach

The Mapper performs a Map-side Join using the department code to obtain the department head and emits the head with a count of 1. The Reducer sums the admissions for each head and identifies the head with the highest admissions.

OUTPUT:

```
root@2f9231d56341:/tmp# hdfs dfs -cat /user/medtrack/output/problem4/part-r-00000
2026-09-20 16:59:06,003 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteHostTrusted = false
Dr. Priya Nair 42
root@2f9231d56341:/tmp#
```

Problem 5 – Month-wise Hospital Admissions

Objective

Calculate the total number of hospital admissions for each month.

Approach

The Mapper extracts the month from the admission date and emits the month with a count of 1. The Reducer sums the counts for each month.

OUTPUT:

```
hadoop@namenode:/tmp$ hdfs dfs -cat /hospital/output/problem5/part-r-00000
2026-01 7
2026-02 8
2026-03 10
2026-04 5
```

YARN Resource Manager Screenshot:

Cluster

About

Nodes

Node Labels

Applications

NEW

NEW SAVING

SUBMITTED

ACCEPTED

RUNNING

FINISHED

FAILED

KILLED

Scheduler

Tools

Cluster Metrics

Apps Submitted	Apps Pending	Apps Running	Apps Completed	Containers Running	Memory Used	Memory Total	Memory Reserved
0	0	0	0	0	0 B	16 GB	0 B

Cluster Nodes Metrics

Active Nodes	Decommissioning Nodes	Decommissioned Nodes	Lost Nodes	Unhealthy Nodes
1	0	0	0	0

Scheduler Metrics

Scheduler Type	Scheduling Resource Type	Minimum Allocation	Maximum Allocation
Capacity Scheduler	[memory-mb (unit=Mi), vcores]	<memory:1024, vCores:1>	<memory:8192, vCores:4>

Show 20 entries

ID	User	Name	Application Type	Queue	Application Priority	StartTime	LaunchTime	FinishTime	State	FinalStatus	Running Containers	Allocated CPU VCores	Allocated Memory MB	Re...
application_1790006778350_0006	root	MedTrack Analytics - P5	MAPREDUCE	default	0	Mon Sep 21 23:10:40 +0550 2026	Mon Sep 21 23:10:40 +0550 2026	Mon Sep 21 23:10:53 +0550 2026	FINISHED	SUCCEEDED	N/A	N/A	N/A	N/...
application_1790006778350_0005	root	MedTrack Analytics - P4	MAPREDUCE	default	0	Mon Sep 21 23:07:45 +0550 2026	Mon Sep 21 23:07:45 +0550 2026	Mon Sep 21 23:07:59 +0550 2026	FINISHED	SUCCEEDED	N/A	N/A	N/A	N/...
application_1790006778350_0004	root	MedTrack Analytics - P1	MAPREDUCE	default	0	Mon Sep 21 23:03:26 +0550 2026	Mon Sep 21 23:03:27 +0550 2026	Mon Sep 21 23:03:40 +0550 2026	FINISHED	SUCCEEDED	N/A	N/A	N/A	N/...
application_1790006778350_0003	root	MedTrack Analytics - P2	MAPREDUCE	default	0	Mon Sep 21 22:56:23 +0550 2026	Mon Sep 21 22:56:24 +0550 2026	Mon Sep 21 22:56:39 +0550 2026	FINISHED	SUCCEEDED	N/A	N/A	N/A	N/...
application_1790006778350_0002	root	MedTrack Analytics - P3	MAPREDUCE	default	0	Mon Sep 21 22:19:01 +0550 2026	Mon Sep 21 22:19:03 +0550 2026	Mon Sep 21 22:19:14 +0550 2026	FINISHED	SUCCEEDED	N/A	N/A	N/A	N/...

Hive VS Pig

Pig	Hive
Procedural Data Flow Language	Declarative SQLish Language
For Programming	For creating reports
Mainly used by Researchers and Programmers	Mainly used by Data Analysts
Operates on the client side of a cluster.	Operates on the server side of a cluster.
Does not have a dedicated metadata database.	Makes use of exact variation of dedicated SQL DDL language by defining tables beforehand.
Pig is SQL like but varies to a great extent.	Directly leverages SQL and is easy to learn for database experts.
Pig supports Avro file format.	Hive does not support it.

Conclusion:

The MedTrack project successfully demonstrated the use of Hadoop MapReduce for analyzing hospital admission data. The five business problems were implemented using Mapper and Reducer components, with Map-side Join used where department information was required. The data was stored and processed through HDFS, and the jobs were executed successfully using YARN.

The project also explored Hive and Pig for re-implementing the average length-of-stay analysis. Overall, the project provided practical experience in distributed data processing, HDFS data management, MapReduce programming, and higher-level big-data tools.